

BGP in IPv4

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Purpose

In this lab, I aimed to learn about BGP, another routing protocol, and try to figure out how to set it up and connect BGP areas with OSPF and EIGRP areas so they could all communicate in IPv4.

Background Information

Introduction Summary on BGP

BGP or the Border Gateway Protocol was first conceived in January 1989 by Yakov Rekhter and Kirk Lougheed. BGP lies on top of TCP and is a Layer 4 protocol that is less complex than OSPF since TCP takes care of some of the elements for routing. It is used to exchange routing information between routers that lie in different autonomous systems (AS). The information that BGP includes is the complete route to all of its destinations and uses that information to maintain a database to see how far the networks can reach. It also uses that information to exchange the information with other BGP systems. BGP is a very vital routing protocol as it deals with traffic on the Internet and if configured wrong, can create issues with availability and security (such as hacking) on websites and online services. Now that we know a bit about BGP, including its important role and how it works, let's dig a little deeper.

Autonomous Systems (AS)

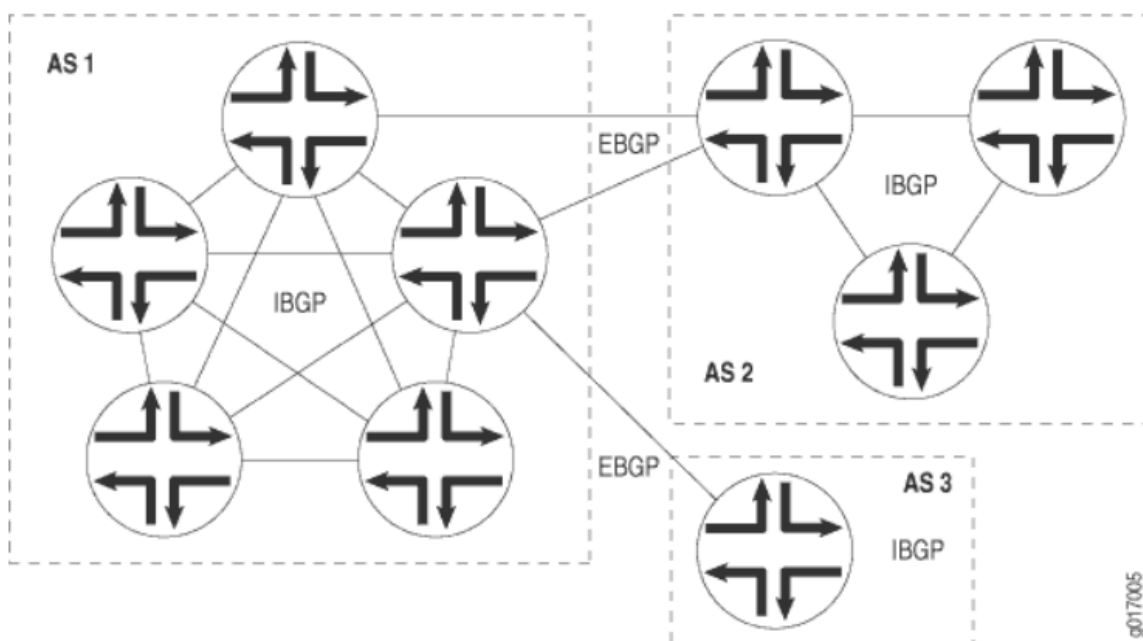
An AS is a collection of Internet Protocol routing prefixes that are connected and controlled by one or multiple network operators under one common administrative entity. Usually you assign a number of your choice to an AS to uniquely identify and distinguish this AS from another AS in the network. In some cases, they need to be the same for different routers to communicate while in others, the numbers need to be different for connections between routers. In routing protocols, it is the routing domain (ex: OSPF area 1 and 2). In terms of its role, BGP will help find a path across the Internet that goes through the smallest number of autonomous systems.

Route Updates

When you boot your devices, the neighbors connected will exchange their entire routing tables. After that, the information exchanged is the routing updates. BGP does not broadcast all of its routing table, differing from RIP, a distance-vector routing protocol that uses the hop count as a routing metric. The route updates are found in a Routing Information Base (RIB), which normally has multiple paths to a destination since routing tables can only hold one route for each destination. In the end, the router is the one who chooses which routes will go to the routing table. After it selects the routes, the paths that will be taken are determined. If by chance, a route is withdrawn, the router can take another one going to the same destination and pick from those in the RIB.

External VS Internal BGP

There are two ways that BGP exchanges routing information: exchanging between different ASs (aka external) VS exchanging inside one AS (aka internal). External BGP uses inter-AS routing while Internal BGP uses intra-AS routing. In External BGP the neighboring devices in a group (also known as external peers) are in different autonomous systems while in Internal BGP groups, all neighboring devices in that group (referred to as internal peers) are within the same AS. There are additional differences between the two types, but I won't go into too much detail as this lab deals mainly with eBGP. Instead, below I inserted a diagram displaying eBGP and iBGP.



This topology shows three autonomous systems and nine routers that are connected. As you can see, there are three iBGPs and two eBGPs connecting between the three IBGPs.

Lab Summary

In this lab I have seven routers and six networks between them with two routers in EIGRP, one in BGP, two in OSPF, and an Area Border Router between EIGRP and BGP and another between BGP and OSPF. Each router was an ISR4321 with a NIM-2T added to each. The routers are connected with Serial DCE cables in Serial interfaces (S0/1/0 and S0/1/1). For the IP addresses, I used addresses from 192.168.1.0 - 192.168.6.0 networks.

Lab Commands

BGP Router-ID #.#.#.#: Statistically sets up the ID for the router in BGP that can be used to identify the network device

Network *specific IP address mask subnet mask*: Identifies which specific network/IP address that the interface will go to

Neighbor *specific IP address remote-as #*: Tells the router in BGP who its neighbors are (neighboring or nearby networks)

Redistribute BGP *AS #*: Imports routes from other routing protocols into BGP so that these routers using different protocols can communicate

Redistribute EIGRP *AS #*: Imports routes from other routing protocols into EIGRP so that these routers using different protocols can communicate

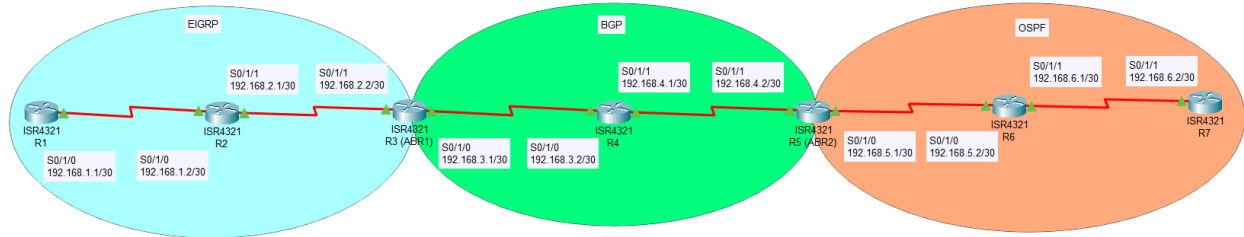
Redistribute OSPF *AS #*: Imports routes from other routing protocols into OSPF so that these routers using different protocols can communicate

Router BGP #: Activates the BGP protocol on the router

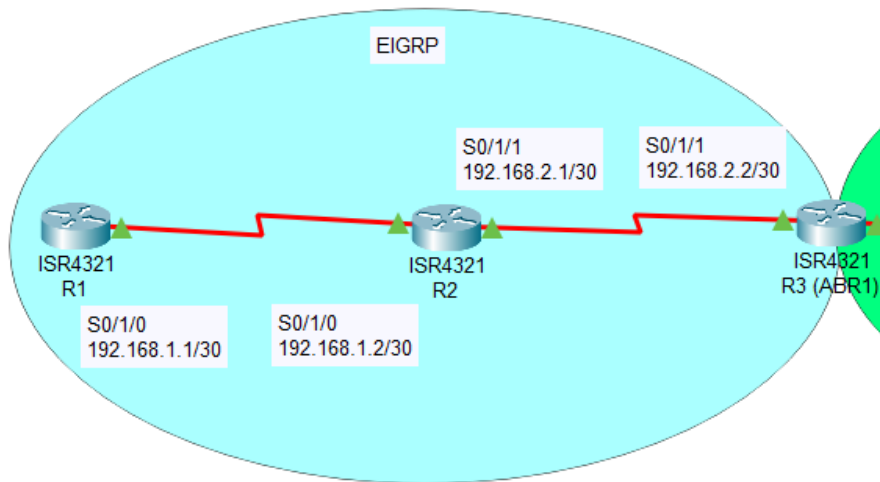
Show IP BGP: Generates the BGP routing table and all its entries

Show IP BGP Summary: Displays the BGP connections' current status

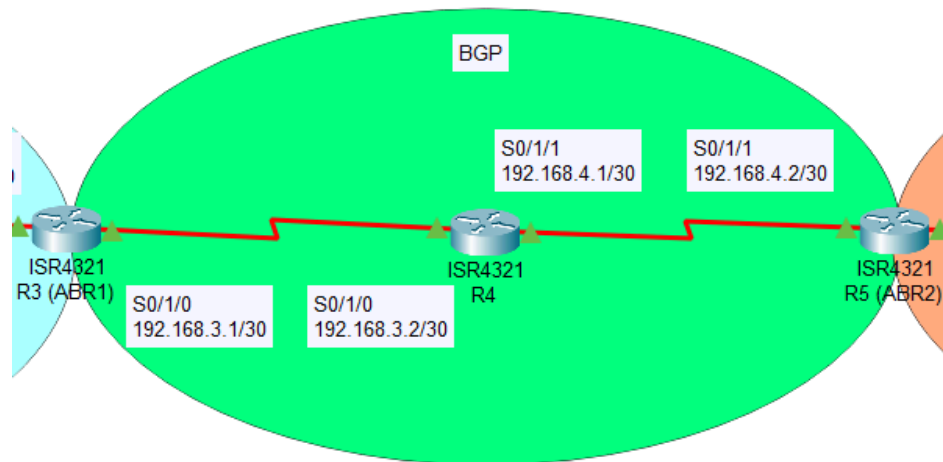
Network Diagram



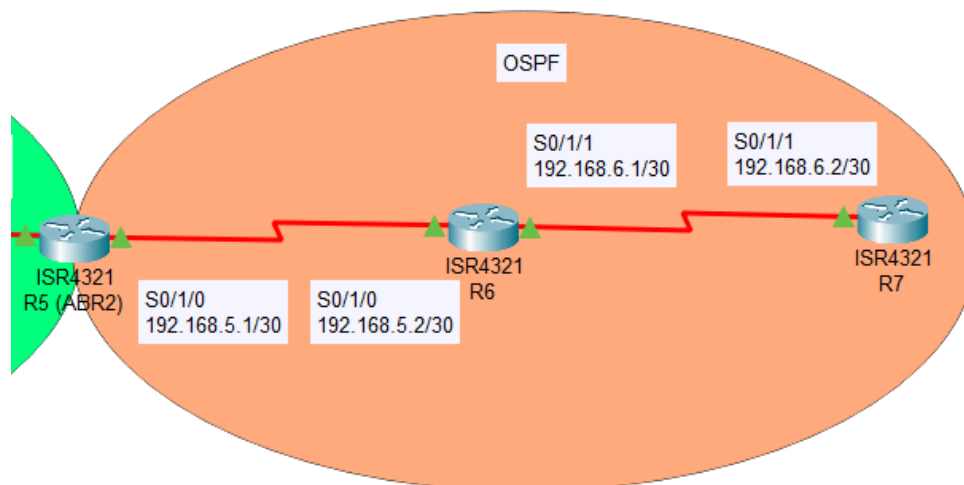
EIGRP



BGP



OSPF



Configurations

Router 1:

- **Show Run**

```
interface Serial0/1/0
 ip address 192.168.1.1 255.255.255.252
 clock rate 2000000

interface Serial0/1/1
 no ip address
```

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```
clock rate 2000000
shutdown
```

```
router eigrp 1
 eigrp router-id 1.1.1.1
 network 192.168.1.0 0.0.0.3
```

- **Show IP Route**

Gateway of last resort is not set

```
192.168.1.0/24 is variably subnetted, 2 subnets, 2
masks
C      192.168.1.0/30 is directly connected, Serial0/1/0
L      192.168.1.1/32 is directly connected, Serial0/1/0
      192.168.2.0/30 is subnetted, 1 subnets
D      192.168.2.0/30 [90/2681856] via 192.168.1.2,
10:11:06, Serial0/1/0
      192.168.4.0/32 is subnetted, 2 subnets
D EX   192.168.4.1/32 [170/26626560] via 192.168.1.2,
01:23:09, Serial0/1/0
D EX   192.168.4.2/32 [170/26626560] via 192.168.1.2,
01:23:09, Serial0/1/0
      192.168.5.0/30 is subnetted, 1 subnets
D EX   192.168.5.0/30 [170/26626560] via 192.168.1.2,
01:23:09, Serial0/1/0
      192.168.6.0/30 is subnetted, 1 subnets
D EX   192.168.6.0/30 [170/26626560] via 192.168.1.2,
01:23:09, Serial0/1/0
```

- **Ping R3 (To BGP)**

- **S0/1/1 192.168.2.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/5/13 ms
```

- **S0/1/0 192.168.3.1**

```
Type escape sequence to abort.
```


Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 2/9/14 ms

- **Ping R4 (To BGP)**

- **S0/1/0 192.168.3.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 3/9/19 ms

- **S0/1/1 192.168.4.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 3/9/20 ms

- **Ping R5 (To BGP)**

- **S0/1/1 192.168.4.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.4.2, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 4/19/58 ms

- **S0/1/0 192.168.5.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.5.1, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 4/13/35 ms

- **Ping R7 (Across the Network)**
 - **S0/1/1 192.168.6.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout
is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 6/20/56 ms
```

Router 2:

- **Show Run**

```
interface Serial0/1/0
 ip address 192.168.1.2 255.255.255.252

interface Serial0/1/1
 ip address 192.168.2.1 255.255.255.252
 clock rate 2000000

router eigrp 1
 eigrp router-id 2.2.2.2
 network 192.168.1.0 0.0.0.3
 network 192.168.2.0 0.0.0.3
```

- **Show IP Route**

```
Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2
    masks
    C       192.168.1.0/30 is directly connected, Serial0/1/0
    L       192.168.1.2/32 is directly connected, Serial0/1/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2
    masks
    C       192.168.2.0/30 is directly connected, Serial0/1/1
    L       192.168.2.1/32 is directly connected, Serial0/1/1
    192.168.4.0/32 is subnetted, 2 subnets
    D EX    192.168.4.1/32 [170/26114560] via 192.168.2.2,
    01:25:26, Serial0/1/1
    D EX    192.168.4.2/32 [170/26114560] via 192.168.2.2,
    01:25:26, Serial0/1/1
```

```
192.168.5.0/30 is subnetted, 1 subnets
D EX    192.168.5.0/30 [170/26114560] via 192.168.2.2,
01:25:26, Serial0/1/1
192.168.6.0/30 is subnetted, 1 subnets
D EX    192.168.6.0/30 [170/26114560] via 192.168.2.2,
01:25:26, Serial0/1/1
```

Router 3 (ABR1):

- **Show Run**

```
interface Serial0/1/0
 ip address 192.168.3.1 255.255.255.252
 clock rate 2000000

interface Serial0/1/1
 ip address 192.168.2.2 255.255.255.252

router eigrp 1
 eigrp router-id 3.3.3.3
 redistribute bgp 1 metric 100 1 255 1 1500
 network 192.168.2.0 0.0.0.3

router bgp 1
 bgp router-id 3.3.3.3
 bgp log-neighbor-changes
 no synchronization
 neighbor 192.168.3.2 remote-as 2
 network 192.168.3.1 mask 255.255.255.255
 redistribute eigrp 1
```

- **Show IP Route**

```
Gateway of last resort is not set

192.168.1.0/30 is subnetted, 1 subnets
D      192.168.1.0/30 [90/2681856] via 192.168.2.1,
07:42:59, Serial0/1/1
192.168.2.0/24 is variably subnetted, 2 subnets, 2
masks
C      192.168.2.0/30 is directly connected, Serial0/1/1
L      192.168.2.2/32 is directly connected, Serial0/1/1
```

```

192.168.3.0/24 is variably subnetted, 2 subnets, 2
masks
C      192.168.3.0/30 is directly connected, Serial0/1/0
L      192.168.3.1/32 is directly connected, Serial0/1/0
      192.168.4.0/32 is subnetted, 2 subnets
B      192.168.4.1/32 [20/0] via 192.168.3.2, 00:00:00
B      192.168.4.2/32 [20/0] via 192.168.3.2, 00:00:00
      192.168.5.0/30 is subnetted, 1 subnets
B      192.168.5.0/30 [20/0] via 192.168.3.2, 00:00:00
      192.168.6.0/30 is subnetted, 1 subnets
B      192.168.6.0/30 [20/0] via 192.168.3.2, 00:00:00

```

- **Show IP BGP**

Network	Next Hop	Metric	LocPrf
*> 192.168.1.0/30	192.168.2.1	0	0 0
1 ?			
*> 192.168.2.0/30	0.0.0.0	0	0 32768
i			
*> 192.168.3.1/32	0.0.0.0	0	0 32768
i			
*> 192.168.4.1/32	192.168.3.2	0	0 0
2 i			
*> 192.168.4.2/32	192.168.3.2	0	0 0
2 3 i			
*> 192.168.5.0/30	192.168.3.2	0	0 0
2 3 ?			
*> 192.168.6.0/30	192.168.3.2	0	0 0
2 3 ?			

- **Show IP BGP Summary**

```

BGP router identifier 3.3.3.3, local AS number 1
BGP table version is 24, main routing table version 6
7 network entries using 924 bytes of memory
7 path entries using 364 bytes of memory
4/4 BGP path/bestpath attribute entries using 736 bytes of
memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32
bytes of memory

```

BGP using 2128 total bytes of memory
BGP activity 7/0 prefixes, 7/0 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
Up/Down	State/PfxRcd						
192.168.3.2	4	2	407	382	24	0	0
04:12:06	4						

- **Ping R1 (Across the Network)**

- **S0/1/0 192.168.1.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/3/10 ms

- **Ping R4 (To BGP)**

- **S0/1/0 192.168.3.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/2/9 ms

- **S0/1/1 192.168.4.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/5/18 ms

- **Ping R5 (To BGP)**

- **S0/1/1 192.168.4.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.2, timeout
is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/8/21 ms

- o **S0/1/0 192.168.5.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.5.1, timeout
is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/5/15 ms

- **Ping R7 (Across the Network)**

- o **S0/1/1 192.168.6.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout
is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/12/22 ms

Router 4:

- **Show Run**

```
interface Serial0/1/0
 ip address 192.168.3.2 255.255.255.252
```

```
interface Serial0/1/1
 ip address 192.168.4.1 255.255.255.252
 clock rate 2000000
```

```
router bgp 2
 bgp router-id 4.4.4.4
 bgp log-neighbor-changes
 no synchronization
 neighbor 192.168.3.1 remote-as 1
 neighbor 192.168.4.2 remote-as 3
 network 192.168.4.1 mask 255.255.255.255
 network 192.168.3.2 mask 255.255.255.255
```

- **Show IP Route**

Gateway of last resort is not set

```

    192.168.1.0/30 is subnetted, 1 subnets
B       192.168.1.0/30 [20/2681856] via 192.168.3.1,
00:00:00
    192.168.2.0/30 is subnetted, 1 subnets
B       192.168.2.0/30 [20/2169856] via 192.168.3.1,
00:00:00
    192.168.3.0/24 is variably subnetted, 3 subnets, 2
masks
C       192.168.3.0/30 is directly connected, Serial0/1/0
B       192.168.3.1/32 [20/0] via 192.168.3.1, 00:00:00
L       192.168.3.2/32 is directly connected, Serial0/1/0
    192.168.4.0/24 is variably subnetted, 3 subnets, 2
masks
C       192.168.4.0/30 is directly connected, Serial0/1/1
L       192.168.4.1/32 is directly connected, Serial0/1/1
B       192.168.4.2/32 [20/0] via 192.168.4.2, 00:00:00
    192.168.5.0/30 is subnetted, 1 subnets
B       192.168.5.0/30 [20/20] via 192.168.4.2, 00:00:00
    192.168.6.0/30 is subnetted, 1 subnets
B       192.168.6.0/30 [20/128] via 192.168.4.2, 00:00:00

```

- **Show IP BGP**

Network	Next Hop	Metric	LocPrf
*> 192.168.1.0/30	192.168.3.1	0	0 0
1 ?			
*> 192.168.2.0/30	192.168.3.1	0	0 0
1 ?			
*> 192.168.3.1/32	192.168.3.1	0	0 0
1 i			
*> 192.168.3.2/32	0.0.0.0	0	0 32768
i			
*	192.168.3.1	0	0 0
1 i			
*> 192.168.4.1/32	0.0.0.0	0	0 32768
i			
*	192.168.4.2	0	0 0
3 i			

```
*> 192.168.4.2/32      192.168.4.2          0      0      0
3 i
*> 192.168.5.0/30      192.168.4.2          0      0      0
3 ?
*> 192.168.6.0/30      192.168.4.2          0      0      0
3 ?
```

- **Show IP BGP Summary**

```
BGP router identifier 4.4.4.4, local AS number 2
BGP table version is 23, main routing table version 6
10 network entries using 1320 bytes of memory
10 path entries using 520 bytes of memory
8/6 BGP path/bestpath attribute entries using 1288 bytes of
memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32
bytes of memory
BGP using 3232 total bytes of memory
BGP activity 8/0 prefixes, 10/0 paths, scan interval 60
secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
Up/Down	State/PfxRcd						
192.168.3.1	4	1	386	388	23	0	0
04:13:45	4						
192.168.4.2	4	3	389	379	23	0	0
05:14:51	4						

- **Ping R1 (Across the Network)**

- **S0/1/0 192.168.1.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/7/18 ms
```

- **Ping R3 (To BGP)**

- **S0/1/1 192.168.2.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/3/14 ms

- **S0/1/0 192.168.3.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/7/31 ms

- **Ping R5 (To BGP)**

- **S0/1/1 192.168.4.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/7/25 ms

- **S0/1/0 192.168.5.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.5.1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/2/6 ms

- **Ping R7 (Across the Network)**

- **S0/1/1 192.168.6.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout
is 2 seconds:
!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 3/8/21 ms

Router 5 (ABR2):

- **Show Run**

```
interface Serial0/1/0
 ip address 192.168.5.1 255.255.255.252
 clock rate 2000000

interface Serial0/1/1
 ip address 192.168.4.2 255.255.255.252

router ospf 1
 router-id 5.5.5.5
 log-adjacency-changes
 redistribute bgp 3 subnets
 network 192.168.5.0 0.0.0.3 area 1

router bgp 3
 bgp router-id 5.5.5.5
 bgp log-neighbor-changes
 no synchronization
 neighbor 192.168.4.1 remote-as 2
 network 192.168.4.2 mask 255.255.255.255
 redistribute ospf 1
```

- **Show IP Route**

```
Gateway of last resort is not set

    192.168.1.0/30 is subnetted, 1 subnets
B       192.168.1.0/30 [20/0] via 192.168.4.1, 00:00:00
    192.168.2.0/30 is subnetted, 1 subnets
B       192.168.2.0/30 [20/0] via 192.168.4.1, 00:00:00
    192.168.3.0/32 is subnetted, 2 subnets
B       192.168.3.1/32 [20/0] via 192.168.4.1, 00:00:00
B       192.168.3.2/32 [20/0] via 192.168.4.1, 00:00:00
    192.168.4.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.4.0/30 is directly connected, Serial0/1/1
L       192.168.4.2/32 is directly connected, Serial0/1/1
```

```

192.168.5.0/24 is variably subnetted, 2 subnets, 2
masks
C      192.168.5.0/30 is directly connected, Serial0/1/0
L      192.168.5.1/32 is directly connected, Serial0/1/0
      192.168.6.0/30 is subnetted, 1 subnets
O      192.168.6.0/30 [110/128] via 192.168.5.2, 10:17:30,
Serial0/1/0

```

- **Show IP BGP**

Network	Next Hop	Metric	LocPrf
*> 192.168.1.0/30	192.168.4.1	0	0 0
2 1 ?			
*> 192.168.2.0/30	192.168.4.1	0	0 0
2 1 ?			
*> 192.168.3.1/32	192.168.4.1	0	0 0
2 1 i			
*> 192.168.3.2/32	192.168.4.1	0	0 0
2 i			
*> 192.168.4.2/32	0.0.0.0	0	0 32768
i			
* 192.168.5.0/30	192.168.5.0	0	0 0
3 ?			
*>	0.0.0.0	0	0 32768
i			
*> 192.168.6.0/30	192.168.5.2	0	0 0
3 ?			

- **Show IP BGP Summary**

```

BGP router identifier 5.5.5.5, local AS number 3
BGP table version is 22, main routing table version 6
8 network entries using 1056 bytes of memory
8 path entries using 416 bytes of memory
4/4 BGP path/bestpath attribute entries using 736 bytes of
memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32
bytes of memory
BGP using 2312 total bytes of memory
BGP activity 7/0 prefixes, 8/0 paths, scan interval 60 secs

```

Neighbor Up/Down	V State/PfxRcd	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
192.168.4.1	4	2	403	384	22	0	0
05:18:41	4						

- **Ping R1 (Across the Network)**

- **S0/1/0 192.168.1.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/12/23 ms
```

- **Ping R3 (To BGP)**

- **S0/1/1 192.168.2.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/8/19 ms
```

- **S0/1/0 192.168.3.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/6/24 ms
```

- **Ping R4 (To BGP)**

- **S0/1/0 192.168.3.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout
is 2 seconds:
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip  
min/avg/max = 1/3/12 ms
```

- o **S0/1/1 192.168.4.1**

```
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout  
is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip  
min/avg/max = 1/4/7 ms
```

- **Ping R7 (Across the Network)**
S0/1/1 192.168.6.2

```
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout  
is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip  
min/avg/max = 2/8/13 ms
```

Router 6:

- **Show Run**

```
interface Serial0/1/0  
 ip address 192.168.5.2 255.255.255.252  
  
interface Serial0/1/1  
 ip address 192.168.6.1 255.255.255.252  
 clock rate 2000000  
  
router ospf 1  
 router-id 6.6.6.6  
 log-adjacency-changes  
 network 192.168.5.0 0.0.0.3 area 1  
 network 192.168.6.0 0.0.0.3 area 1
```

- **Show IP Route**

```
Gateway of last resort is not set
```

```

    192.168.1.0/30 is subnetted, 1 subnets
O E2    192.168.1.0/30 [110/20] via 192.168.5.1, 04:19:56,
Serial0/1/0
    192.168.2.0/30 is subnetted, 1 subnets
O E2    192.168.2.0/30 [110/20] via 192.168.5.1, 04:19:56,
Serial0/1/0
    192.168.3.0/32 is subnetted, 2 subnets
O E2    192.168.3.1/32 [110/20] via 192.168.5.1, 03:26:17,
Serial0/1/0
O E2    192.168.3.2/32 [110/20] via 192.168.5.1, 03:49:09,
Serial0/1/0
    192.168.4.0/32 is subnetted, 1 subnets
O E2    192.168.4.2/32 [110/20] via 192.168.5.1, 03:26:43,
Serial0/1/0
    192.168.5.0/24 is variably subnetted, 2 subnets, 2
masks
C        192.168.5.0/30 is directly connected, Serial0/1/0
L        192.168.5.2/32 is directly connected, Serial0/1/0
    192.168.6.0/24 is variably subnetted, 2 subnets, 2
masks
C        192.168.6.0/30 is directly connected, Serial0/1/1
L        192.168.6.1/32 is directly connected, Serial0/1/1

```

Router 7:

- **Show Run**

```

interface Serial0/1/0
  no ip address
  clock rate 2000000
  shutdown
!
interface Serial0/1/1
  ip address 192.168.6.2 255.255.255.252
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 1
  router-id 7.7.7.7
  log-adjacency-changes
  network 192.168.6.0 0.0.0.3 area 1

```

- **Show IP Route**

Gateway of last resort is not set

```
192.168.1.0/30 is subnetted, 1 subnets
O E2    192.168.1.0/30 [110/20] via 192.168.6.1, 04:22:14,
Serial0/1/1
192.168.2.0/30 is subnetted, 1 subnets
O E2    192.168.2.0/30 [110/20] via 192.168.6.1, 04:22:14,
Serial0/1/1
192.168.3.0/32 is subnetted, 2 subnets
O E2    192.168.3.1/32 [110/20] via 192.168.6.1, 03:28:35,
Serial0/1/1
O E2    192.168.3.2/32 [110/20] via 192.168.6.1, 03:51:27,
Serial0/1/1
192.168.4.0/32 is subnetted, 1 subnets
O E2    192.168.4.2/32 [110/20] via 192.168.6.1, 03:29:01,
Serial0/1/1
192.168.5.0/30 is subnetted, 1 subnets
O       192.168.5.0/30 [110/128] via 192.168.6.1, 10:24:59,
Serial0/1/1
192.168.6.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.6.0/30 is directly connected, Serial0/1/1
L       192.168.6.2/32 is directly connected, Serial0/1/1
```

- **Ping R1 (Across the Network)**

- **S0/1/0 192.168.1.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 6/10/17 ms
```

- **Ping R3 (To BGP)**

- **S0/1/1 192.168.2.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout
is 2 seconds:
!!!!!!
```

Success rate is 100 percent (5/5), round-trip
min/avg/max = 6/10/17 ms

- **S0/1/0 192.168.3.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/12/24 ms

- **Ping R4 (To BGP)**

- **S0/1/0 192.168.3.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 3/9/23 ms

- **S0/1/1 192.168.4.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 3/8/14 ms

- **Ping R5 (To BGP)**

- **S0/1/1 192.168.4.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/7/16 ms

- **S0/1/0 192.168.5.1**

Type escape sequence to abort.


```
Sending 5, 100-byte ICMP Echos to 192.168.5.1, timeout
is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 2/8/23 ms
```

Problems

One issue I had was figuring out what I am supposed to put in my redistribute commands (for all three routing protocols). At this point, I didn't quite understand what the numbers and other information coming after "redistribute (routing protocol) #" meant and what I was supposed to put there. After looking online and discussing with friends, I discovered how to configure these commands.

Another challenge I faced was that I couldn't ping across the BGP routers. After checking my configurations, I noticed I wrote the wrong network statements for all the BGP routers (R3-5). I wrote the IP address of the neighbor, not the address that is configured on the router. Re-configuring this allowed me to ping across the BGP routers.

After resolving the BGP network statements everything could ping but both R1 and R7 could not ping past 192.168.2.2. To resolve this, I checked my configurations and my routes using the "show run" and "show IP route" commands. Finally, after double checking multiple times, I discovered that I wrote the redistribute command incorrectly on my ABR1 router (R3). I forgot to write numbers for the metric part of the command and after fixing it, I was able to ping all routers to all the other routers.

Conclusion

Through this lab, connecting EIGRP, BGP, and OSPF routers gave me a deeper understanding of working with different routing protocols and how to connect them so these different networks can communicate. Specifically, I learned about BGP and understood how redistribute commands work better. Also, I had another opportunity to work with a larger network than I am used to and have increased my drive to know more about networking through

my own research on BGP and the history of it. Most importantly, this lab taught me to be careful when typing in my configurations because every little mistake can make a big difference.